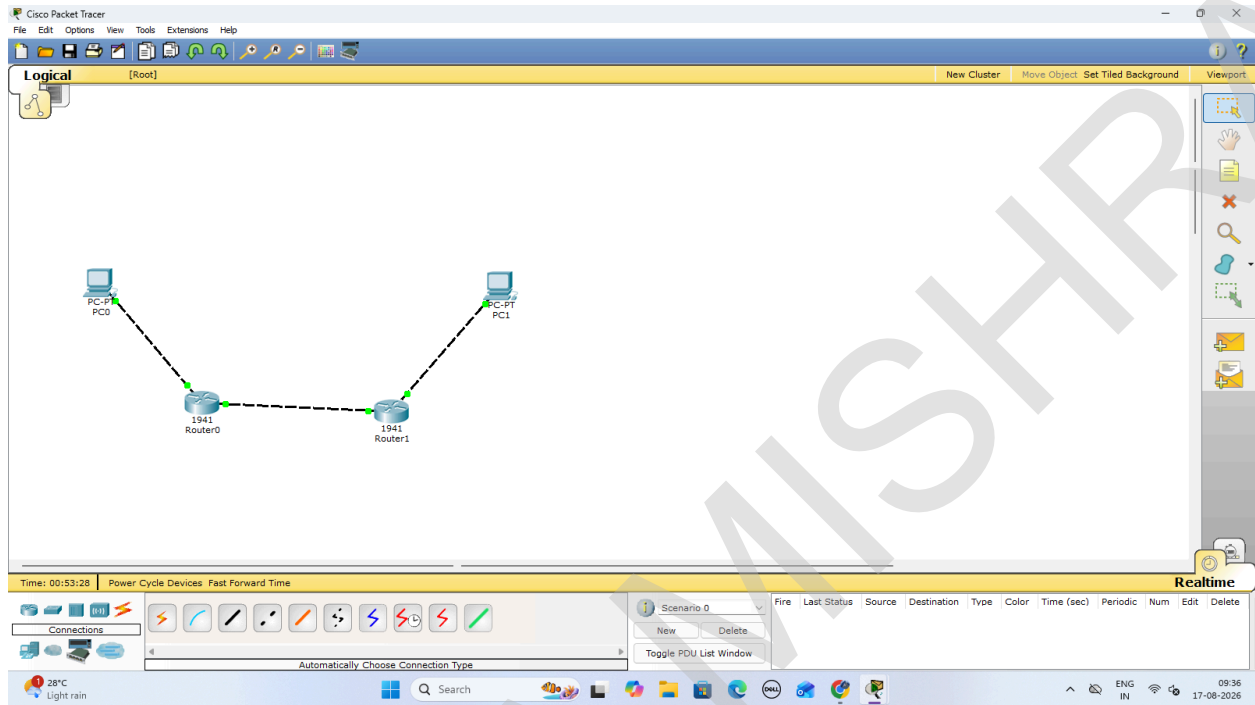


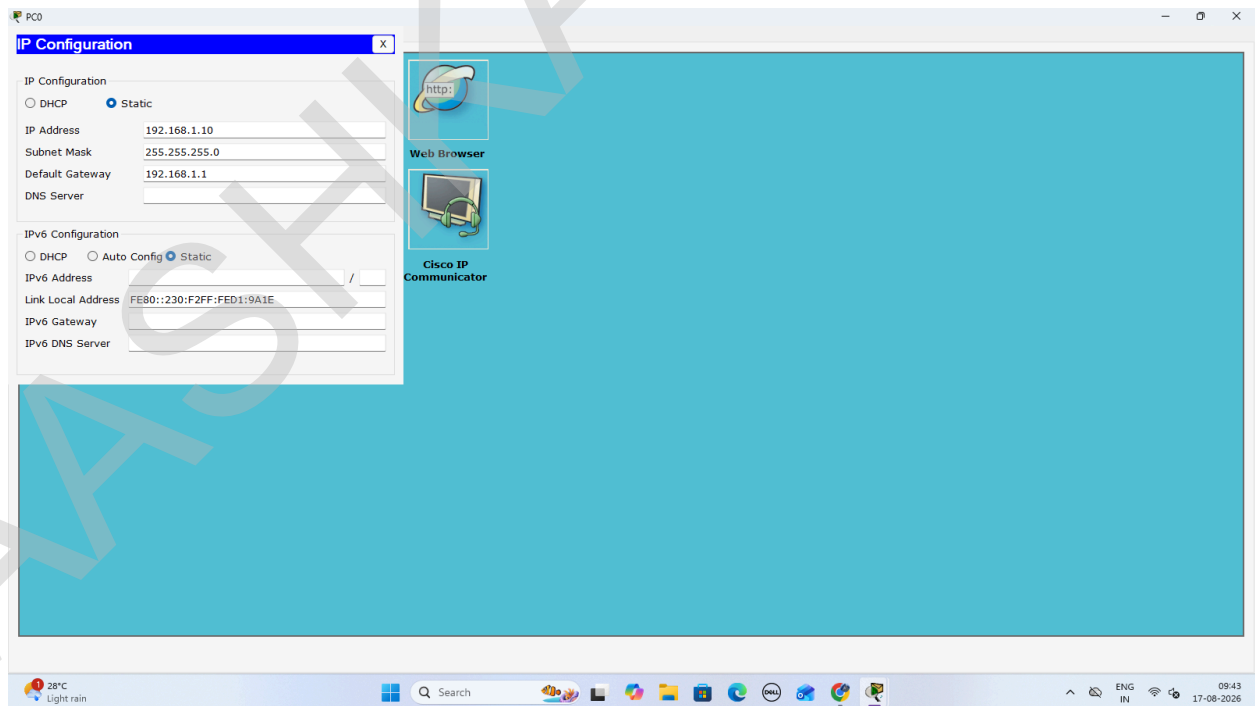
PRACTICAL NO. 6

AIM: Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

CONNECTION:



PC 0 -

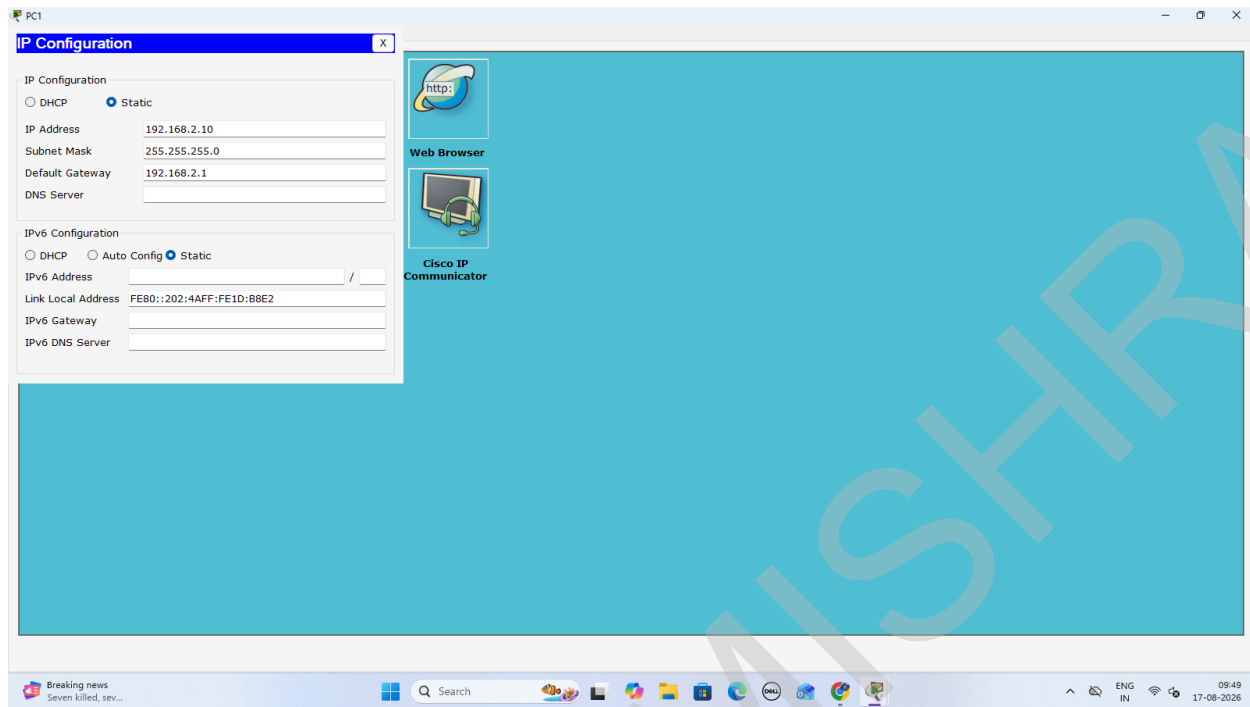


NAME: AASHKA MISHRA

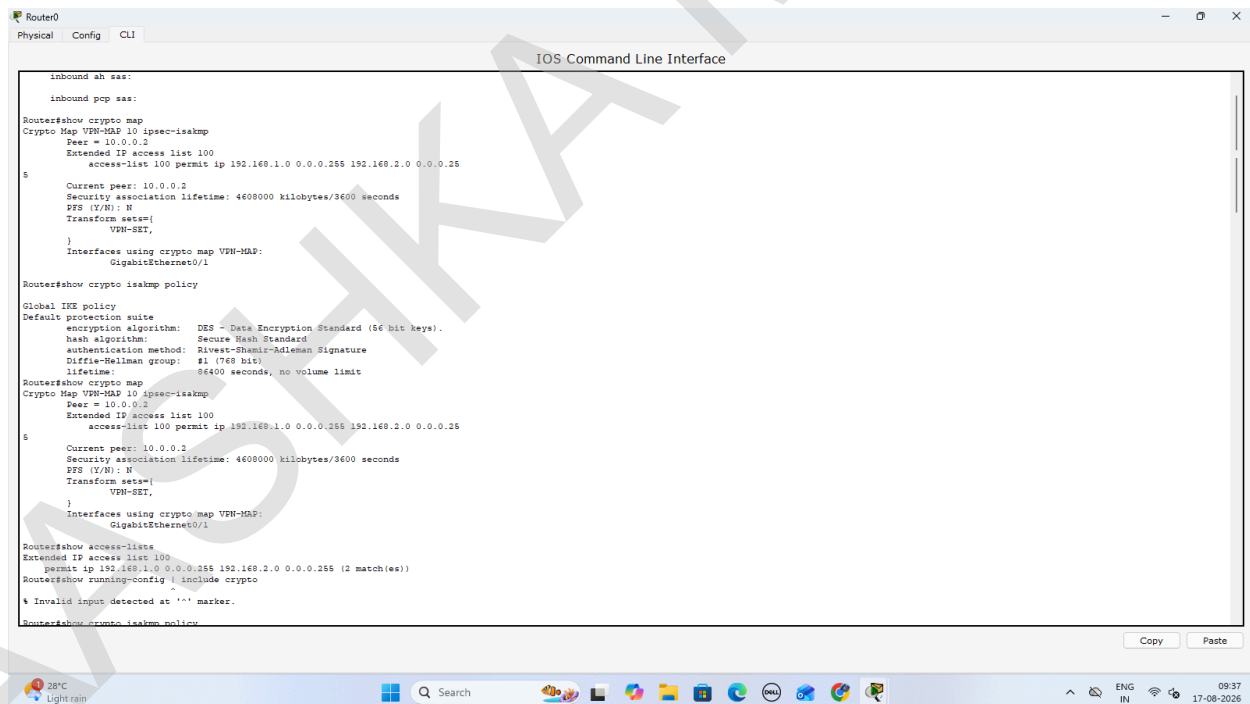
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: CYBER SECURITY

PC 1 -



ROUTER 0 -



NAME: AASHKA MISHRA
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE

SUBJECT: CYBER SECURITY

```
Router0
Physical Config CLI

IOS Command Line Interface

Router#show access-lists
Extended IP access list 100
 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 (2 match(es))
Router#show running-config | include crypto
-
% Invalid input detected at '^' marker.
Router#show crypto isakmp policy
Global IKE policy
Default protection suite
 encryption algorithm: DES - Data Encryption Standard (56 bit keys)
 hash algorithm: Secure Hash Standard
 authentication method: Rivest-Shamir-Adleman Signature
 Diffie-Hellman group: #1 (768 bit)
 lifetime:
Router#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
 peer = 10.0.0.2
 Extended IP access list 100
 access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.25
5
 Current peer: 10.0.0.2
 Security association lifetime: 4608000 kilobytes/3600 seconds
 PFS (Y/N) : N
 Transform sets=
 VPN-SET,
 Interfaces using crypto map VPN-MAP:
 GigabitEthernet0/1
Router#show access-lists
Extended IP access list 100
 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 (2 match(es))
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.2
A pre-shared key for address mask 10.0.0.2 255.255.255.255 already exists!
Router(config)#end
%SYS-6-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.1.1 YES manual up up
GigabitEthernet0/1 10.0.0.1 YES manual up up
Vlan1 unassigned YES NVRAM administratively down down
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/30 is directly connected, GigabitEthernet0/1
L 10.0.0.1/32 is directly connected, GigabitEthernet0/1
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
L 192.168.1.1/32 is directly connected, GigabitEthernet0/0
S 192.168.2.0/24 (1/0) via 10.0.0.2
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/30 is directly connected, GigabitEthernet0/1
L 10.0.0.1/32 is directly connected, GigabitEthernet0/1
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
L 192.168.1.1/32 is directly connected, GigabitEthernet0/0
S 192.168.2.0/24 (1/0) via 10.0.0.2
Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst src state conn-id slot status
10.0.0.2 10.0.0.1 QM_IDLE 1085 0 ACTIVE
```

NAME: AASHKA MISHRA
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: CYBER SECURITY

```
Router0
Physical Config CLI

IOS Command Line Interface

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   10.0.0.0/30 is directly connected, GigabitEthernet0/1
L   10.0.0.1/32 is directly connected, GigabitEthernet0/1
C   192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.1.0/24 is directly connected, GigabitEthernet0/0
L   192.168.1.1/32 is directly connected, GigabitEthernet0/0
S   192.168.2.0/24 (1/0) via 10.0.0.2

Router#show crypto isakmp sa
dst          src          state      conn-id slot status
10.0.0.2     10.0.0.1     QM_IDLE    1086     0 ACTIVE

IPv6 Crypto ISAKMP SA

Router#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPM-MAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
#pkts decaps: 6, #pkts decrypt: 6, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0x471F1F1E(1193221918)

inbound esp sas:
spi: 0x10ED7C7C(275610748)
transform: esp-aes esp-sha-hmac ,

Router#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPM-MAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
```

```
Router0
Physical Config CLI

IOS Command Line Interface

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
#pkts decaps: 6, #pkts decrypt: 6, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0x471F1F1E(1193221918)

inbound esp sas:
spi: 0x10ED7C7C(275610748)

Router#show crypto isakmp sa
IPv6 Crypto ISAKMP SA
dst          src          state      conn-id slot status
10.0.0.2     10.0.0.1     QM_IDLE    1086     0 ACTIVE

IPv6 Crypto ISAKMP SA

Router#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPM-MAP, local addr 10.0.0.1

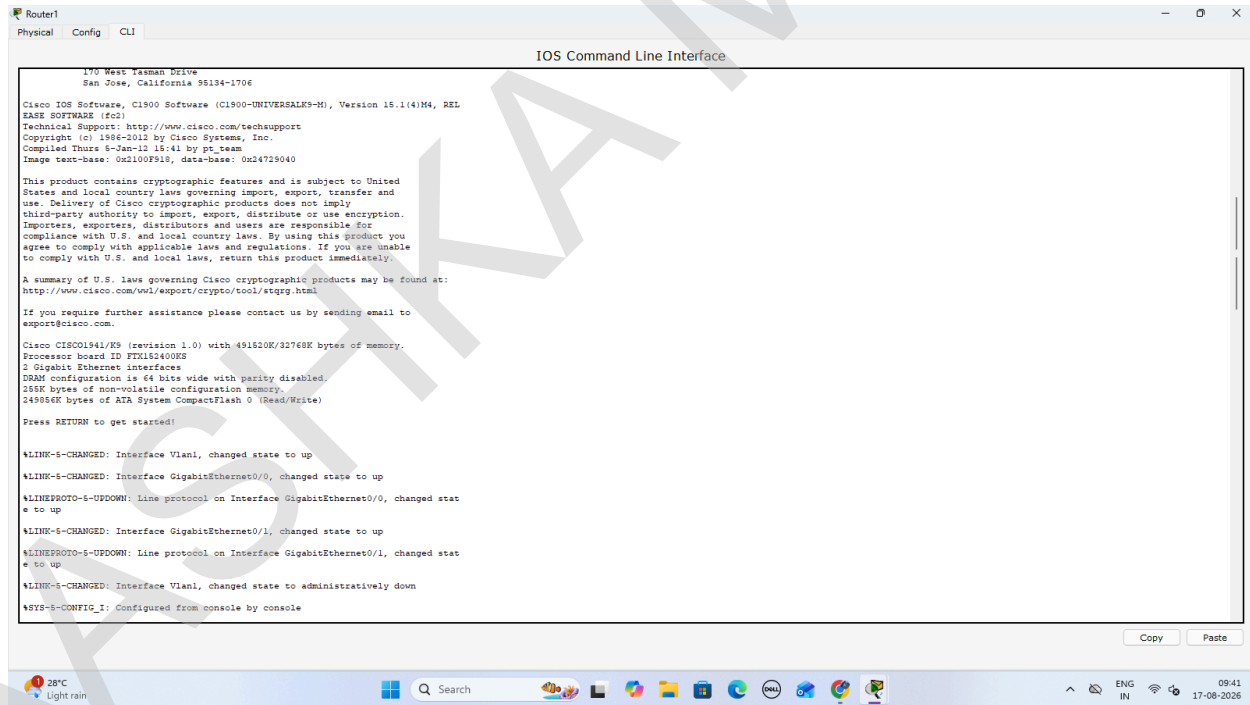
protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 11, #pkts encrypt: 11, #pkts digest: 0
#pkts decaps: 10, #pkts decrypt: 10, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0x471F1F1E(1193221918)

inbound esp sas:
spi: 0x10ED7C7C(275610748)
--More--
```

NAME: AASHKA MISHRA
ROLL NO: T095

ROUTER 1 -



NAME: AASHKA MISHRA
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE

SUBJECT: CYBER SECURITY

```
Router1
Physical Config CLI
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed stat
e to up
%LINK-5-CHANGED: Interface Vlan1, changed state to administratively down
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed stat
e to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed stat
e to up

Router(config)#crypto isakmp 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0
.255
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#crypto map VPN-MAP
*Jan 3 07:16:24.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show crypto ipsec sa
Crypto Map VPN-MAP 10 ipsec-isakmp
Peer = 10.0.0.1
Extended IP access list 100
access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.25
5
Current peer: 10.0.0.1
Security association lifetime: 4608000 kilobytes/3600 seconds
PFS (Y/N): N
Transform sets=
VPN-SET,
Interfaces using crypto map VPN-MAP:
GigabitEthernet0/1

Router#show crypto ipsec sa
Interface: GigabitEthernet0/1
Crypto map tag: VPN-MAP, local addr 10.0.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.2.0/255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.1.0/255.255.0/0/0)
current peer 10.0.0.1 port 800
PERMIT, flags=(origin_is_acl,)
#pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

local crypto endpt.: 10.0.0.2, remote crypto endpt.: 10.0.0.1
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0x0(0)

inbound esp sas:
inbound ah sas:

Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst src state conn-id slot status
IPv4 Crypto ISAKMP SA

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
```

NAME: AASHKA MISHRA
ROLL NO: T095

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```
Router1
Physical Config CLI
IOS Command Line Interface

inbound esp sas:
inbound ah sas:

Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst      src      state      conn-id slot status
IPv6 Crypto ISAKMP SA

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#exit
Router(config)#
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
A pre-shared key for address mask 10.0.0.1 255.255.255.255 already exists!
Router(config)#end
*Aug-1-CONFIG_I: Configured from console by console

Router#

Router con0 is now available

Press RETURN to get started.
```

PC 0 -

```
PC0
Physical Config Desktop Custom Interface
Command Prompt

Packet Tracer PC Command Line 1.0
PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>ping 192.168.2.10

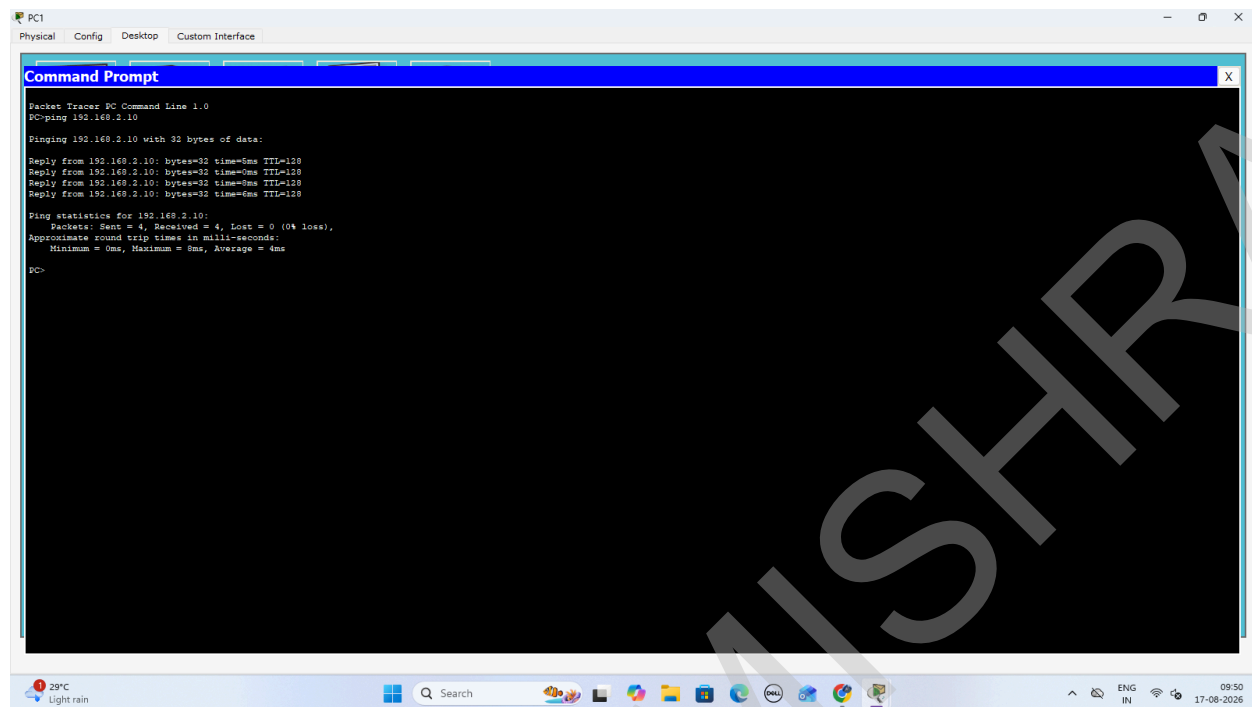
Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
```

NAME: AASHKA MISHRA
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: CYBER SECURITY

PC 1 -



The screenshot shows a Packet Tracer PC Command Line window for PC1. The window title is "Command Prompt". The text inside the window is as follows:

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=0ms TTL=128
Reply from 192.168.2.10: bytes=32 time=0ms TTL=128
Reply from 192.168.2.10: bytes=32 time=0ms TTL=128
Reply from 192.168.2.10: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 4ms

PC>
```

The window is part of a larger application with tabs for "Physical", "Config", "Desktop", and "Custom Interface". The "Desktop" tab is selected. The taskbar at the bottom shows the Windows logo, a search bar, and various application icons. The system tray on the right shows the date and time as 09:50 on 17-08-2026.

NAME: AASHKA MISHRA
ROLL NO: T095